

Viraj Sanghavi

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Education

Dwarkadas J. Sanghvi College of Engineering, B.Tech in Computer Engineering – Mumbai, India Sept 2024 – present
CGPA: 9.15

Experience

Developer Intern, ITAakash Strategic Software Pvt. Ltd. – Mumbai, India June 2026 – Aug 2026

- Built an AI cash-flow forecasting system using Amazon Chronos-2, replacing a legacy XGBoost pipeline, predicting daily net cash flow across thousands of grouped financial records
- Designed a Flask API serving 365-day forecasts per (project, ledger, company) group, integrated into the existing Java EE ERP via PostgreSQL
- Implemented walk-forward backtesting (5 rolling origins, 4-month horizon) with bias correction for right-skewed construction cash flow data

Projects

Vigilia - Graph-Native AML Detection Jan 2025 – present

- Started as Argus at iDEA 2.0 (Union Bank of India PS3, Team Zeta, Top 40+ finalist); continued past the hackathon under the name Vigilia
- Built graph-native AML system detecting 8 structural typologies plus 7 behavioural signals (structuring, smurfing, layering) on 500K+ IBM synthetic transactions, rebuilt since on a Multi-GNN (PNACConv + GINEConv) classifying transactions directly on the multigraph
- Added FastAPI + PostgreSQL persistence with a documented schema-migration process, an audit trail, and a pytest suite covering pure decision logic

Hermes - Adaptive ASR Pipeline Jan 2025 – present

- Built a research-grade adaptive ASR pipeline: Wav2Vec2 transcription, 8-feature acoustic noise fingerprinting across 5 environment categories, and TTS-driven closed-loop remediation
- Implemented a phoneme-level drift monitor (20-utterance rolling window) plus 5+ published techniques including AdaLoRA (Zhang et al., ICLR 2023) and HDMoLE LoRA-MoE routing (Mu et al., ICASSP 2025)
- Deployed full-stack system (FastAPI, React, PostgreSQL, Docker) with CI/CD via GitHub Actions on Render

Fortuna - Algorithmic Trading System Jan 2025 – present

- Built a stacked RF/XGBoost/LightGBM ensemble with regime routing across 20+ equities and crypto pairs, backtested via walk-forward optimization (Sharpe up to 3.46), with SHAP/LIME explainability, drift monitoring, and a 144-test suite

Research

Publicity & Research Member, DJS Anveshan - Research Club – Dwarkadas J. Sanghvi College of Engineering Jan 2024 – present

- Authored a problem discovery report on a critical Federated Learning failure mode: robust aggregation filters (Krum, FLAME) silently discarding valid non-IID updates from minority institutions in cross-institutional fraud detection
- Identified an unaddressed gap in FL security literature as of 2025, citing IEEE Transactions on Big Data and JPMorgan's Project Aikya

Skills

Languages / Infra: Python, SQL, TypeScript, Bash, C++, Git, Docker, FastAPI, React, PostgreSQL, n8n

ML / Data: PyTorch, PyTorch Geometric, scikit-learn, XGBoost, LightGBM, NumPy, Pandas, Hugging Face